

Muhammad Ibrahim

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EDUCATION

New York University <i>Bachelor of Science in General Engineering, Mechanical and Civil Focus</i> - Major GPA - 3.60 / 4.00, UK (2:1). - Major Coursework - Solid Mechanics, Engineering Dynamics, Fluid Mechanics, Thermodynamics, Heat Transport, Composite Materials.	Abu Dhabi, UAE May 2026
New York University <i>Study Away, Immigration and Theatre</i> Analysed migration in French theatre casting and the economics of integration; Term GPA - 3.70 / 4.00.	Paris, France Jan 2024
New York University June Term <i>Study Away, Architecture and Climate Impact</i> Examined how architectural decisions shape climate outcomes and purpose-led design; Term GPA - 4.00 / 4.00.	Valletta, Malta Jun 2023

EXPERIENCE

Center for Stability, Instability and Turbulence (SITE) <i>Graduate Intern, New York University</i> - Delivered the model's entire training dataset solo in 6 weeks, 60 trials across 4 angles of attack. - Tripled the model's valid range to 45 degrees from a 15 degree dataset, grounding it in dimensional analysis. - Eliminated 50+ further tunnel runs, with a model trained in Python and MATLAB to predict untested angles.	New York, USA May 2026 - Jul 2026
Advanced Manufacturing Workshop <i>Workshop Technician, New York University</i> - Cut print failure rates from 50% to 10% across over 70 print hours by diagnosing root causes and refining parameters. - Raised part strength 50% over plain printed filament, laying up bidirectional carbon fibre and fibreglass.	Abu Dhabi, UAE Sep 2025 - May 2026
Center for Stability, Instability and Turbulence (SITE) <i>Undergraduate Research Assistant, New York University</i> - Generated a 130+ case Rayleigh-Taylor and Richtmyer-Meshkov dataset in a 2-person team, now under post-processing. - Achieved boundary-layer separation in every run across 5 water-tunnel model variants, verified by flow visualisation.	Abu Dhabi, UAE Apr 2023 - May 2026
Energy and Propulsion Lab <i>Undergraduate Research Assistant, New York University</i> - Boosted upper-level ventilation 28% for AeroFarms growing towers, simulating airflow in OpenFOAM and Converge. - Presented Urban Air Mobility propulsion findings to a 25-member Hyundai NGV panel in South Korea.	Abu Dhabi, UAE Nov 2022 - Nov 2023

PROJECTS

Fixed-Wing UAV for Search and Rescue Payload Deployment, Personal Project <i>Jun 2026 - Present</i> - Drove a 1,130 mm 3D-printed airframe and 6-part powertrain from concept to finished prototype. - Reduced airframe mass from 2500 to 1500 g via lightweight PLA, infill tuning, internal geometry and carbon tubing. - Engineered an airborne release for a 600 g multirotor daughter craft, delivered as a prototype for flight testing.	<i>Jun 2026 - Present</i>
Advanced Anti-Roll Tank System (ART3S), Mubadala Investment and Abu Dhabi Maritime Academy <i>Oct 2024 - Sep 2025</i> - Damped vessel roll amplitude 85% passively, engineering 200 to 800 m³ anti-heeling tanks for ferry and car-carrier hulls. - Shortened roll decay time 33% passively and 62% under active control, modelling vessel dynamics in Simulink.	<i>Oct 2024 - Sep 2025</i>
Savonius Turbine for Traffic Wind Energy Harvesting, Capstone Design <i>Sep 2024 - May 2025</i> - Raised peak moment 72% over the plain baseline, adding serrated edges and a 45-degree helical twist in ANSYS Fluent. - Increased power 56% over the non-serrated baseline in CFD, adding serrated leading edges. - Proved roadside self-start within 2 to 5 seconds on a 1:3 carbon fibre rotor, projecting 3 to 15 W at full scale.	<i>Sep 2024 - May 2025</i>
Compressor-Less Multi-Stage TEC Refrigerator, New York University <i>Mar 2025 - May 2025</i> - Cooled a cabin from 27 to 10 degrees in 24 minutes without a compressor, running 2 thermoelectric modules in parallel. - Held the 10 degree setpoint to within half a degree, the sensor's limit, tuning a closed-loop PID controller on Arduino.	<i>Mar 2025 - May 2025</i>
Formula Student SAE Drivetrain Design, New York University <i>Sep 2024 - May 2025</i> - Exposed a half-shaft 49% over its twist limit before build, by simulating shock loads that static checks had missed. - Redirected the redesign to geometry, saving the cost of an alloy with zero twist gain, by proving stiffness ignores yield.	<i>Sep 2024 - May 2025</i>

AWARDS

Research Grant, Mubadala and Abu Dhabi Maritime Association	<i>Aug 2024</i>
Industry Recognition and Grant, Hyundai NGV	<i>Nov 2022</i>
Full Merit Scholarship, New York University	<i>Dec 2021</i>

EXTRACURRICULAR ACTIVITIES

NYU Abu Dhabi Athletics, Varsity Jiu-Jitsu, Recreational Sport and Team Videographer <i>2022 - 2026</i> - Competed in jiu-jitsu at university varsity level, earning selection in each of four years. - Filmed and photographed the university cricket team across a full season, delivering footage and stills each week. - Recreational sport across padel, golf, tennis, cricket, kayaking, volleyball, badminton, football, cycling and wall climbing.	<i>2022 - 2026</i>
Habitat for Humanity Nepal, Volunteer, Residential Build Programme <i>2025</i> - Constructed a single-storey house from foundation to roof with a volunteer team, handing a finished home to its family. - Checked site dimensions and layouts against the civil drawings before concrete was poured. - Converted donated labour and materials into permanent housing for families priced out of building.	<i>2025</i>
Emirates Motorsports Organization, Chief Track Marshal <i>2019 - 2025</i> Directed trackside safety and first response across 15 events at Yas Marina Circuit and Dubai Autodrome.	<i>2019 - 2025</i>

SKILLS

CAD and Design - SolidWorks (CSWA), mechanical design, technical drawings, design for manufacturing, rapid prototyping.
Simulation - computational fluid dynamics (CFD), finite element stress analysis (FEA), ANSYS Fluent, Simulink, HPC.
Manufacturing - FDM 3D printing, carbon fibre and fibreglass composites, Arduino, closed-loop PID control.
Programming - Python, MATLAB, machine learning, sensor data acquisition.
Languages - English C2, Urdu C2, Arabic A2.